EEN1095 Project Progress and Change Log

Student Name	Dhanush Kumar Ramesh	**Student ID**	A00013665
Project Title	Yield-Aware Inverse Design of High-Speed Interconnects: A Physics-Constrained Generative Approach	**Supervisor**	Dr. Marissa Condon
Implementation Period	Week 3 - Week 4	**Version Date**	04/06/2026

Guidance: Maintain this as one **cumulative** document throughout EEN1095. Add a new entry every two weeks and do not remove earlier records. Progress should be recorded in a timely and concise manner. Where no change has occurred, this should be stated explicitly.

1. Project Overview

Original approved project title	Yield-Aware Inverse Design of High-Speed Interconnects: A Physics-Constrained Generative Approach
Original project objectives	Build an yield aware inverse model for the high-speed interconnects to predict the geometries from the target s-parameters
Original proposed methodology	Tandem network with forward model as a validator and inverse model predicting geometries with Jacobian regularization for yield optimization
Planned outputs / deliverables	Yield optimized inverse model predicting the geometries from the input s-parameter and global features.

2. Summary of Key Milestones

The table below is intended as a high-level summary. Use Section 3 for the detailed bi-weekly updates.

Period	Planned activity	Progress made	Outcome / status
Weeks 1-2	Complete the Forward model, start working on Inverse design	Forward model completed, inverse design fundamental design steps and execution started	Data Pipeline – completed Forward Model – Completed (partially) Inverse design – In-Progress

Weeks 3-4	Design the baseline architecture for the inverse design model and try implementing yield optimization		In-Progress
Weeks 5-6			[Completed / In progress / Revised]
Weeks 7-8			[Completed / In progress / Revised]
Weeks 9-10			[Completed / In progress / Revised]
Weeks 11-12			[Completed / In progress / Revised]
Weeks 13-14			[Completed / In progress / Revised]

3. Detailed Progress Entries

Entry 1: Weeks 1-2

Main objectives for this period	Complete Forward Model, start inverse design
Work completed	Forward model trained and completed with average MAE of 2.12dB on a complex TUHH dataset
Key outputs produced	Forward model with MAE of 2.12dB tracking the curve.
Problems encountered	Forward Model tracks the resonance dip (struggles a bit on 55GHz and above range)
Changes made to plan or methodology	The Data pipeline logic was changed to extract the 4x4 matrix of all differential pairs to augment the dataset and increase the datapoints available. Changed from rational forward net to 1D CNN – Direct sequence ResNet
Reasons for any changes	Data-pipeline change was to increase the no of datapoints. Forward model change was CNN performed better than Rational Model
Next steps	Working on the inverse model

Entry 2: Weeks 3-4

Main objectives for this period	Design and implement the inverse CVAE architecture in the tandem network with frozen weights from the forward model

Work completed	Implemented tandem network successfully and connected the baseline cvae design to link with the frozen weight from the forward model.
Key outputs produced	The baseline model predicts the geometries from the given s-parameter profile and the forward model validates the generated design to keep the geometries in the desired scope obeying physical constraints.
Problems encountered	Strict penalty functions making the curve to focus on the low frequency range and ignoring the resonance dips in the high frequency range.
Changes made to plan or methodology	No changes in the plan – the baseline CVAE architecture was implemented and the tandem network was connected.
Reasons for any changes	N/A
Next steps	Improvise the inverse model – implement yield optimization and adding the active learning loop to the pipeline.

Entry 3: Weeks 5-6

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 4: Weeks 7-8

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]

Next steps	[Insert concise notes here]

Entry 5: Weeks 9-10

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 6: Weeks 11-12

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]
Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]

Entry 7: Weeks 13-14

Main objectives for this period	[Insert concise notes here]
Work completed	[Insert concise notes here]
Key outputs produced	[Insert concise notes here]
Problems encountered	[Insert concise notes here]
Changes made to plan or methodology	[Insert concise notes here]

Reasons for any changes	[Insert concise notes here]
Next steps	[Insert concise notes here]